

CS3860
Lab Assignment 2

I want you to write a Smalltalk script that will read a text file and generate some statistics on that file. The statistics that I'm looking for are :

- Number of characters, words, and lines
- The frequency of occurrence of each word
- The frequency of occurrence of each letter

For this program, a word is defined to be a character string consisting of upper-case and/or lower-case letters. Words with different cases are considered to be different. Once your program has finished reading the file and collecting the stats, it will display them as well as the histograms. For instance, if your program is given the input "My name is my name and #\$\$%^&* yours is franco is123is" then its output layout should look as shown below. Please note that the output you see below may not be correct; I am simply trying to show you how your output should look like.

Words in dictionary - my,name,is,and,yours,franco

Number of Letters : 23
Number of Words : 17
Number of Lines : 1

```
/-----\  
| Letter Freq |  
\-----/
```

```
a |*****  
b |***** (22)
```

```
.  
.   
.   
y |**  
z |
```

```
/-----\  
| Dictionary |  
\-----/
```

Word	Frequency
my	2
name	2
is	4
and	1
yours	1
franco	1

```

/-----\
| Histogram |
\-----/

          *
          *
        * * *
      * * * * *
-----
          ^ ^ ^ ^ ^ ^
my   -----/ | | | | |
name -----/ | | | | |
is   -----/ | | | | |
and  -----/ | | | | |
yours -----/ | | | | |
franco-----/ | | | | |

```

Note that in the histogram the line of dashes must start at the longest word!! I also want to explain the (22) that appears on the bar for the letter 'b'. If a letter appears more than 10 times then the balance will be displayed at the end of the bar for that letter.

To help you get started, I offer you the following script. I will explain it in the lab.

```

francoc@franco-dell-desktop:~/cs3860$ more diction.st
'Please enter name of file : ' display.
filename := stdin nextLine .
f := FileStream open: filename mode: FileStream read.
fileContents := WriteStream on: (String new).
f linesDo: [:x | x printOn: fileContents ].
words := fileContents contents tokenize: '^[A-Za-z]+'.
"
words displayNl.
words size displayNl.
"
uniq := Set withAll: words.
"
uniq displayNl.
"
freq := Bag withAll: words.
"
freq displayNl.
(String new: 5 withAll: $-) displayNl.
"
sorted := freq sortedByCount.
sorted do: [ :x | ((x value),
                  (String new: ( 20 - (x value size)) withAll: $-),
                  (String new: (x key) withAll: $*)
                ) displayNl].

```

As you can see from my code, you may find yourself using most of the containers.